

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 9- Hypergeometric Distribution

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Session Outline

- 1 Experiment of Hypergeometric Variable
- 2 Moments
- 3 Applications

Session Overview

In this session, students are introduced to Hypergeometric probability distribution. The properties of the hypergeometric distribution will be discussed. Computation of the probabilities using the hypergeometric probability mass function will be presented.

Learning Outcomes

At the end of the session, students will be able to:

- 1 Describe the experiment leading to the hypergeometric random variable.
- 2 State the probability mass function of the hypergeometric distribution and show that it is a probability function.
- 3 State the expectation and variance of hypergeometric distribution.
- 4 Compute probabilities and moments using the hypergeometric distribution function.

Experiment of Hypergeometric Random Variable

- Suppose that a box contains b blue marbles and r red marbles.
- Let us perform n trials of an experiment in which a marble is chosen at random, its colour is observed, and then the marble is put back in the box.
- This type of experiment is often referred to as sampling with replacement.
- In such a case, if X is the random variable denoting the number of blue marbles chosen (successes) in n trials, then X will follow the binomial distribution.
- If we modify the experiment such that, the chosen marbles are not replaced after being chosen, then the sampling (experiment) is without replacement.

Hypergeometric Distribution

- Then X is a random variable having the Hypergeometric distribution with parameters n, b and $b + r$.
- Hence,

$$P(X = x) = \frac{\binom{b}{x} \binom{r}{n-x}}{\binom{b+r}{n}}, \quad \text{where } x = 0, 1, 2, \dots, n \text{ \& } n \leq b$$

- Saying that:

$\binom{b}{x}$: is the number of ways of selecting x blue marbles from b marbles.

Hypergeometric Distribution

- $\binom{r}{n-x}$: is the number of ways of selecting $n-x$ red marbles from r marbles.
- $\binom{b+r}{n}$: is the number of ways of selecting any n marbles from $b+r$ marbles.
- We denote the above probability mass function as:

$X \sim H(n, b, N)$, where n is the sample size,
 b is the number of success items and
 $N = b + r$, the population size.

Legitimacy Condition

Theorem 1

Given that $X \sim H(n, b, N)$ show that, where $N = b + r$

$$P(X = x) = \frac{\binom{b}{x} \binom{r}{n-x}}{\binom{N}{n}}, \quad n \leq N; \quad n \leq b, \quad x = 0, 1, 2, \dots, n$$

is a probability function.

Proof

- 1 It is clear that $\binom{b}{x} > 0, \forall x, b; \binom{r}{n-x} > 0, \forall r \geq 0$ and $\binom{N}{n} > 0, \forall n, N. \implies P(X = x) \geq 0, \forall x.$

Legitimacy Proof

Proof Cont'd

$$\text{ii } \sum_{\text{all } x} P(X = x) = 1;$$

$$\Rightarrow \sum_{x=0}^n P(X = x) = \sum_{x=0}^n \frac{\binom{b}{x} \binom{r}{n-x}}{\binom{N}{n}}$$

Now by using the Chu-Vandermode identity, which states that

$$\binom{a+b}{k} = \sum_{x=0}^n \binom{a}{x} \binom{b}{k-x}$$

Legitimacy Proof

Proof Cont'd.

$$\sum_{x=0}^n P(X=x) = \sum_{x=0}^n \frac{\binom{b}{x} \binom{r}{n-x}}{\binom{N}{n}} = \frac{\binom{b+r}{n}}{\binom{N}{n}} = \frac{\binom{N}{n}}{\binom{N}{n}} = 1$$

Hence the proof.

Theorem 2:

Given that $X \sim H(n, b, N)$ show that, where $N = b + r$, then

$$E(X) = np; \quad \text{Var}(X) = npq \left(\frac{N - n}{N - 1} \right),$$

where $p = \frac{b}{N}$ and $q = 1 - p$.

Sketch Proof:

The mean and variance for this distribution are

$$E(X) = \frac{nb}{b + r}; \quad \text{Var}(X) = \frac{nbr(b + r - n)}{(b + r)^2(b + r - 1)}$$

Sketch Proof:

If we let the total number of blue and red marbles be N , while the proportions of blue and red marbles are p and $q = 1 - p$, respectively, then

Sketch Proof Cont'd:

$$p = \frac{b}{b+r} = \frac{b}{N}, \quad q = \frac{r}{b+r} = \frac{r}{N} \text{ or } b = Np, \quad r = Nq$$

so that we have $P(X = x)$ and moments become, respectively,

$$P(X = x) = \frac{\binom{Np}{x} \binom{Nq}{n-x}}{\binom{N}{n}}$$

$$E(X) = np \quad \text{and} \quad \text{Var}(X) = npq \left(\frac{N-n}{N-1} \right)$$

Example 1

A box contains 6 red and 14 white marbles of the same size. An experiment consists of drawing 5 marbles from the box. Let X represents the number of white marbles drawn. Write down without proof the probability distribution of $(X = x)$, if the selection is one at a time (include in each case the range of values of x)

- a without replacement
- b with replacement
- c Evaluate $P(X \geq 2)$ under (a) and (b).

Solution

Given that $n = 5$, $b = 14$, white marbles and $r = 6$, red marbles. Therefore, $N = 20$, the total marbles in the box.

- a The probability distribution of X is the Hypergeometric distribution and is given as

$$P(X = x) = \frac{\binom{14}{x} \binom{6}{5-x}}{\binom{20}{5}}, \quad x = 0, 1, \dots, 5$$

$$\begin{aligned} P(X \geq 2) &= 1 - P(X \leq 1) = 1 - \left[\frac{\binom{14}{0} \binom{6}{5}}{\binom{20}{5}} + \frac{\binom{14}{1} \binom{6}{4}}{\binom{20}{5}} \right] \\ &= 1 - \frac{9}{646} = 0.986 \end{aligned}$$

- b The probability distribution of X is the Binomial distribution and is given as

$$P(X = x) = \binom{5}{x} \left(\frac{7}{10}\right)^x \left(\frac{3}{10}\right)^{5-x}, \quad x = 0, 1, \dots, 5$$

$$\begin{aligned} P(X \geq 2) &= 1 - P(X \leq 1) = 1 - \left[\binom{5}{0} \left(\frac{7}{10}\right)^0 \left(\frac{3}{10}\right)^5 + \binom{5}{1} \left(\frac{7}{10}\right)^1 \left(\frac{3}{10}\right)^4 \right] \\ &= 1 - \frac{1539}{50000} = 0.969 \end{aligned}$$

Example 2

A box contains 6 blue marbles and 4 red marbles. An experiment is performed in which a marble is chosen at random and its colour observed, but the marble is not replaced. Find the probability that after 5 trials of the experiment, 3 blue marbles will have been chosen.

Solution

Given that, $n = 5$, $b = 6$, $r = 4$, $N = 10$ and $x = 3$, we have

$$P(X = 3) = \frac{\binom{6}{3} \binom{4}{2}}{\binom{10}{5}} = \frac{10}{21}$$

Binomial Approximation to Hypergeometric

Given that $X \sim H(n, b, N)$ with N large, then

$$\lim_{N \rightarrow \infty} P(X = x) = \lim_{N \rightarrow \infty} \frac{\binom{b}{x} \binom{r}{n-x}}{\binom{N}{n}} \\ \approx \binom{n}{x} p^x q^{n-x},$$

where $p = \frac{b}{N}$ and $q = 1 - p$.

Note: The proof is left as an exercise for students.

Some comments on the above results:

- $\text{Var}(X) \rightarrow npq$ as $N \rightarrow \infty$, since $\lim_{N \rightarrow \infty} \left(\frac{N-n}{N-1} \right) = 1$.
- Also, the above results show that for sufficiently large N , the distribution of X may be approximated by the Binomial distribution.

Question 1

A voter's register consists of 6000 females and 4000 males. Ten persons are selected at random to form a committee. Let X be the number of females selected. Determine the probability that more males than females are on the committee.

Solution

Since N is large, it implies that $X \sim b(n, p)$, where $n = 10$ and $p = \frac{6000}{10000} = 0.6$

$$P(X = x) = \binom{10}{x} (0.6)^x (0.4)^{10-x}, \quad x = 0, 1, 2, \dots, 10$$

The required probability is $P(X < 5)$.

$$P(X < 5) = \sum_{x=0}^4 \binom{10}{x} (0.6)^x (0.4)^{10-x} = 0.166$$

THANK YOU!!!